Building a personality prediction model for social media users from the MBTI dataset

Lê Vũ Khánh Thảo, Hoàng Ngọc Quý, Trần Hoàng Nguyên, Nguyễn Đức Thịnh, Hoàng Anh Dũng
ThS. Huỳnh Văn Tín, ThS. Nguyễn Văn Kiệt
Faculty of Information Science and Engineering, University of Information Technology,
Ho Chi Minh City, Vietnam
Vietnam National University, Ho Chi Minh City, Vietnam
{23521469, 21522529, 21522396, 21522636, 20521206 }gm.uit.edu.vn

Abstract

In the digital era, the linguistic traces on social networks are a valuable source of information reflecting human psychological characteristics. This report focuses on researching the application of Natural Language Processing (NLP) and Machine Learning techniques to predict the four dichotomous personality pairs of the MBTI model from text data. Through experimenting with and comparing various methods—ranging from traditional models such as logistic regression and SVM to modern deep learning architectures such as LSTM, TextCNN, and particularly the large language model RoBERTa—the research team has evaluated in detail the performance, contextual understanding capabilities, and challenges of processing imbalanced real-world data. The research results not only contribute scientifically to the field of social media data mining but also open up practical application potential in human resource recruitment, empathy marketing, and personalized user experiences.

1 Introduction

1.1 Background

In the digital era, social networks have become a place where people express their thoughts, emotions, and personal behaviors through text. These digital footprints contain valuable information about users' psychological traits. Automatic personality recognition from online text is attracting significant attention from researchers aiming to develop systems that truly understand users.

However, traditional personality assessment methods (such as psychological questionnaires) are often costly and difficult to implement on a large scale. Therefore, applying Natural Language Processing (NLP) and Machine Learning techniques to predict personality from social media data is an urgent and promising direction.

1.2 Objective

The project focuses on solving the multi-label text classification problem within the context of noisy and imbalanced social media data. Specific objectives include:

- Building and optimizing classification models: Designing and training a variety of machine learning models, ranging from traditional methods such as Logistic Regression and SVM combined with TF-IDF to modern deep learning models like LSTM and CNN using Word Embedding techniques inherited from previous research. The team will further train the dataset with more advanced NLP models such as RoBERTa (Transformer-based). The objective is to identify the most optimal architecture for recognizing the four pairs of dichotomous personality traits in the MBTI model (I-E, N-S, T-F, J-P) from natural language text.

- Addressing data challenges: Focus on handling core issues of real-world datasets, such as severe class imbalance between personality groups (e.g., the large discrepancy between Intuition and Sensing groups) and the data leakage phenomenon commonly found in self-reported datasets.

- Experimental comparative evaluation: Conduct a quantitative comparison between the performance of simple models (Explainable AI) and complex models (Black-box models) on a small-scale dataset. Through this, the research aims to address the trade-off between computational cost, complexity, and practical accuracy in the personality prediction problem.

1.3 Significance

- Scientific Significance: The research contributes to the fields of Social Media Mining and Natural Language Processing (NLP) by:

+ Demonstrating the feasibility of predicting implicit personality traits through linguistic footprints, rather than relying on traditional psychological tests.

+ Providing empirical evaluations of the limitations of Large Language Models (LLMs) when applied to specialized datasets with small sizes and high noise levels.

- Practical Significance: The model's results have the potential for widespread application in various fields of life and business:

+ Automated Recruitment: Assisting businesses in preliminary candidate screening based on cultural and personality fit, thereby saving recruitment time and costs.

+ Personalizing User Experience: Serving as a foundation for building smarter Recommender Systems that suggest content, books, movies, or products tailored to each individual's psychological tendencies.

+ Empathic Marketing (Psychographic Marketing): Helping marketers gain deeper insights into target customers not only through demographics but also through psychographics, thereby optimizing advertising messages to achieve higher engagement efficiency.

2 Theoretical Framework

2.1 Overview of MBTI (Myers-Briggs Type Indicator))

2.1.1 Concept and Origin

MBTI (Myers-Briggs Type Indicator) is a psychological assessment tool developed by Katharine Cook Briggs and her daughter, Isabel Briggs Myers, based on the theoretical foundations of Carl Gustav Jung's analytical psychology. It is one of the world's most popular instruments used in human resource management, education, and personal development to classify human personalities.

2.1.2 The Four Personality Dichotomies

MBTI classifies personality based on an individual's natural preferences across four dichotomous axes, forming 16 distinct personality groups:

- Introversion (I) - Extraversion (E): Energy orientation. Introverts (I) focus on their inner world, while extroverts (E) orient toward the outer world and social interaction.

- Intuition (N) - Sensing (S): Information perception. The Intuition (N) group tends toward theories and abstract models, while the Sensing (S) group relies on reality and observable details.

- Thinking (T) - Feeling (F): Decision making. The Thinking (T) group relies on logic and objective rules; conversely, the Feeling (F) group prioritizes personal values and empathy.

- Judging (J) - Perceiving (P): Approach to the world. The Judging (J) group prefers planning and order, while the Perceiving (P) group favors spontaneity and adaptability to circumstances.

2.2 Natural Language Processing (NLP)

2.2.1 Basic Concepts and Characteristics of Social Media Data

Natural Language Processing (NLP) is a branch of artificial intelligence focused on the interaction between computers and human language. In the context of personality classification from social media, the data is typically characterized by its unstructured nature, containing significant noise (teencode, abbreviations, emojis), and a lack of grammatical standards. Processing this data requires robust preprocessing techniques and models capable of capturing flexible semantic meanings.

2.2.2 Text Representation Methods

In order for computers to understand language, text must be converted into numerical vectors. The research team utilizes the following methods:

- TF-IDF (Term Frequency-Inverse Document Frequency): This is a statistical method for evaluating the importance of a word within a document. The weight of a word increases proportionally to the number of times it appears in that document but is inversely proportional to its frequency across the entire dataset. Advantages: Highly effective at filtering out common, low-meaning words (stopwords) and highlighting specific keywords characteristic of each personality group.

- Word Embeddings (GloVe & FastText): Representing words as dense vectors in a multidimensional space, where words with similar meanings are positioned close to each other. FastText: Unlike GloVe, which learns from whole words, FastText learns representations based on character n-grams (sub-word information). This enables FastText to effectively handle out-of-vocabulary (OOV) words and slang variations commonly found on social media.

- Contextual Embeddings (Transformer-based): Unlike static Word Embeddings (where each word has a single fixed vector), models like BERT/RoBERTa generate dynamic representation vectors based on the specific context of the word within a sentence, helping to resolve issues with polysemy (words with multiple meanings).

2.2.3 Classification Models

The research team applied and compared three main groups of models:

- Linear Models (Logistic Regression / SVM):

+ These are traditional machine learning models commonly used as baselines.

+ Logistic Regression: Models the probability of a sample belonging to a certain class. When combined with TF-IDF (sparse data), Logistic Regression operates very effectively, with high speed and ease of interpretation.

+ SVM (Support Vector Machine): Finding an optimal hyperplane to separate data classes in a multi-dimensional space.

- Deep Learning (TextCNN & LSTM):

+ TextCNN (Convolutional Neural Networks for Text): Using convolutional filters with different window sizes to extract important local features (n-grams) within the text.

+ LSTM (Long Short-Term Memory): A variant of RNN capable of remembering long-term dependencies in text sequences. LSTM addresses the vanishing gradient problem of traditional RNNs, making it suitable for processing long posts.

- Transformer & RoBERTa:

+ Transformer: Architecture based entirely on the Self-Attention mechanism, allowing the model to process the entire text sequence in parallel and capture complex semantic relationships.

+ RoBERTa (Robustly Optimized BERT Approach): An improved version of BERT. RoBERTa removes the Next Sentence Prediction (NSP) task, utilizes a larger volume of training data, and employs dynamic masking techniques to achieve superior performance in language understanding tasks. In this study, RoBERTa is used to extract deep contextual features for the personality classification problem.

3 Research Methodology

3.1 Dataset Description: mbti_1/ MBTI 500

This research was conducted by simultaneously examining two datasets of different scales to comprehensively evaluate model performance, from basic to advanced.

The first is the baseline dataset (mbti_1), collected from the PersonalityCafe forum with a scale of 8,675 samples. Each sample consists of the user's last 50 posts. However, this dataset faces significant limitations regarding size and severe class imbalance (the Intuition/Sensing ratio is as high as 7:1), which poses challenges for traditional machine learning tasks.

To overcome the aforementioned limitations, the study utilizes an expanded dataset (MBTI 500) compiled from Reddit and PersonalityCafe, featuring over 106,000 records (12 times larger than the previous dataset). The most significant improvement of MBTI 500 is the standardization of each data sample to a fixed length of 500 words. This structure not only ensures sufficient context to reveal deep psychological traits but also perfectly aligns with the input limits of modern Transformer models (such as BERT, RoBERTa), thereby optimizing the feature extraction process.

Both datasets reflect the common challenge of Class Imbalance due to self-selection bias, where the Introversion (I) and Intuition (N) groups account for an overwhelming proportion compared to the Extroversion (E) and Sensing (S) groups. To address this issue, the study applies specialized processing techniques such as Class Weighting and Focal Loss to ensure fairness during the model training process.

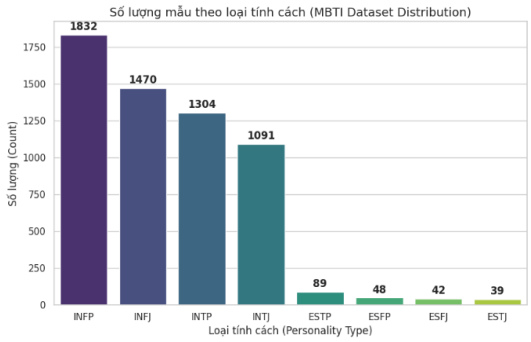

Figure 1: Number of majority and minority type labels in the mbti_1 dataset

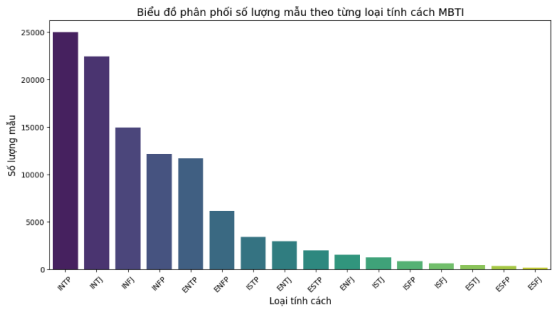

Figure 2: Number of majority and minority type labels in the MBTI 500 dataset

3.2 Preprocessing

To transform raw social media data into clean vectors that a computer can understand, the team applied a preprocessing pipeline consisting of the following steps:

3.2.1 Data Cleaning

Noise Removal: Remove URLs and replace them with a special token [LINK]. Eliminate special characters and emojis that do not carry semantic meaning.

3.2.2 Anti-Data Leakage

This is the most critical step proposed by the team as an enhancement over baseline studies. Since the data is sourced from personality forums, users frequently self-identify their personality types within their posts (e.g., "As an ENTP, I think...").

- Solution: Utilize Regular Expressions (Regex) to scan and completely remove the 16 personality identifier keywords (such as "INTJ", "ENFP", "INTP", etc.) occurring within the post content.

- Purpose: To ensure the model learns genuine linguistic features instead of "memorizing" (overfitting) the labels already present in the text.

3.2.3 Splitting the Prediction Axes

Instead of building a Multi-class Classification model to directly predict 16 personality types (which entails high complexity and low accuracy), the team transformed the problem into four independent Binary Classification tasks corresponding to the four dichotomies:

- Introversion (I) – Extraversion (E)
- Intuition (N) – Sensing (S)
- Thinking (T) – Feeling (F)
- Judging (J) – Perceiving (P)

3.3 Model Design and Training Hyperparameters

The research team conducted a series of experiments ranging from baseline models to advanced architectures to identify the optimal approach for this dataset.

3.3.1 Baseline Model: TF-IDF + LinearSVC

Representation: Use TF-IDF to extract lexical features.

Algorithm: Use LinearSVC (Support Vector Classification). This serves as the baseline model for initial performance evaluation.

Limitations: It fails to effectively address the data imbalance issue, resulting in a low Recall score for the minority classes.

3.3.2 Deep Learning Experiments

To capture deeper semantic meaning, the team experimented with the following architectures:

- LSTM (Long Short-Term Memory): Combined with Word Embeddings (GloVe/FastText) to capture sequential dependencies within the text.

- TextCNN: Utilizes FastText Embeddings combined with convolutional neural networks to extract critical n-grams.

3.3.3 Advanced Experimental Model: RoBERTa (Transformer-based)

Instead of limiting the study to statistical models, the team conducted experiments with the RoBERTa (Robustly Optimized BERT Pretraining Approach) large language model. This effort aims to leverage Deep Contextual Understanding, a capability that TF-IDF models lack.

- Architecture: Utilizing roberta-base as the backbone for feature extraction, connected to a Linear Classifier layer at the output.

- Fine-tuning Strategy: The model is retrained on the mbti_1 dataset with a low learning rate (2e-5) to avoid disrupting pre-trained weights, while Contextual Augmentation (context-based synonym generation) is applied to enrich the data due to the small size of the dataset. For the MBTI 500 dataset, Contextual Augmentation is not used; instead, a "Head + Tail" strategy is applied: extracting the first 200 tokens (introduction/context) and concatenating them with the last 250 tokens (conclusion). The total of 450 tokens ensures the input fits within RoBERTa's 512-token limit while retaining the core content. Additionally, Soft Freezing is applied to the entire model, which is trained with an extremely small learning rate (1e-5) for gentle fine-tuning without breaking previously learned linguistic knowledge.

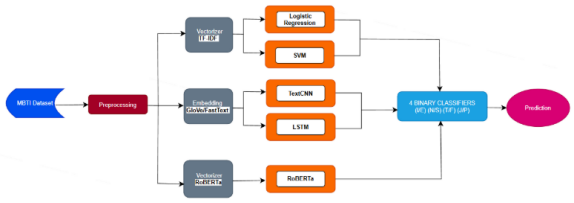

Figure 3: Overview of architecture diagram and training model.

4 Experiments and Results

4.1 Training Results

The research team conducted experiments on three different groups of models to comprehensively evaluate personality classification performance on the MBTI Kaggle dataset.

4.1.1 Traditional Machine Learning Models (Baseline & Comparison)

The team employed classical statistical methods combined with TF-IDF to establish the baseline performance level.

- Baseline: LinearSVC (Support Vector Machine):

Axis	Dataset	Accuracy	MF1	WF1
IE	mbti_1	81.67%	69.00%	80.00%
	MBTI-500	84.27%	80.06%	84.84%
NS	mbti_1	87.32%	60.00%	84.00%
	MBTI-500	91.34%	77.89%	92.14%
TF	mbti_1	79.65%	80.00%	80.00%
	MBTI-500	89.47%	88.56%	89.55%
JP	mbti_1	70.84%	69.00%	70.00%
	MBTI-500	79.72%	79.33%	79.79%

Figure 4: Evaluation of SVM model Performance

+ Impact of Big Data: MBTI-500 helps the algorithm determine the optimal hyperplane and margin, completely overcoming the noise phenomenon of small datasets.

+ N–S Axis (92.14%): Achieved the highest performance thanks to the clear lexical separation between N and S.

+ T–F & J–P Axes: Breakthrough growth ($\sim$ 9˘10%), proving that linear models can fully handle complex axes when provided with sufficient data.

- Baseline: Logistic Regression (Comparison Model):

Axis	Dataset	Accuracy	MF1	WF1
IE	mbti_1	77.12%	68.00%	77.00%
	MBTI-500	88.45%	84.00%	89.00%
NS	mbti_1	85.76%	71.00%	86.00%
	MBTI-500	91.14%	79.00%	92.00%
TF	mbti_1	79.60%	80.00%	80.00%
	MBTI-500	92.81%	86.00%	93.00%
JP	mbti_1	70.78%	70.00%	71.00%
	MBTI-500	82.17%	82.00%	82.00%

Figure 5: Evaluation of Logistic Regression Model Performance

+ Explosive Performance: Accuracy increased by an average of 10–13%, completely overcoming the underfitting phenomenon thanks to large-scale data.

+ Optimal Class Balance: Macro F1 rose significantly (I–E: 68% $\rightarrow$ 84%), demonstrating the outstanding effectiveness of class_weight in recognizing minority groups.

+ High Standard: The most difficult axis, J–P, achieved >82%, establishing a benchmark that challenges Deep Learning models.

4.1.2 Deep Learning Model Group (Experiments and Comparison)

The team experimented with neural network architectures to explore the ability of automatic feature extraction superior to TF-IDF.

- LSTM (Long Short-Term Memory):

Axis	Dataset	Accuracy	Macro Avg F1	Weighted Avg F1
IE	mbti_1	72.05%	65.00%	73.00%
	MBTI-500	79.53%	72.90%	79.16%
NS	mbti_1	84.90%	56.00%	82.00%
	MBTI-500	81.46%	62.20%	84.61%
TF	mbti_1	72.80%	71.00%	71.00%
	MBTI-500	86.08%	73.77%	87.20%
JP	mbti_1	61.21%	41.00%	48.00%
	MBTI-500	71.97%	69.75%	71.48%

Figure 6: Evaluation of LSTM Model Performance

+ Overcoming Mode Collapse (J–P Axis): Macro F1 surged (41% $\rightarrow$ 70%), demonstrating that LSTM requires large-scale data to escape majority-class prediction.

+ Balancing the N–S Axis: Accepted a slight decrease in Accuracy to improve Macro F1 (56% $\rightarrow$ 62%), prioritizing fairness and reducing rote learning of the majority group.

+ Limitation: Overall performance is still lower than the Baseline, as the sequential architecture struggles to retain information in very long texts (500 words).

- TextCNN (Convolutional Neural Networks):

Axis	Dataset	Accuracy	Macro Avg F1	Weighted Avg F1
IE	mbti_1	76.71%	62.00%	75.00%
	MBTI-500	79.00%	71.98%	79.29%
NS	mbti_1	84.27%	59.00%	82.00%
	MBTI-500	88.99%	71.03%	89.88%
TF	mbti_1	72.91%	73.00%	73.00%
	MBTI-500	82.80%	81.46%	82.98%
JP	mbti_1	66.74%	64.00%	66.00%
	MBTI-500	70.08%	69.65%	70.23%

Figure 7: Evaluation of TextCNN Model Performance

+ Advantage of N-gram Extraction: Optimized keyword feature capture on large datasets, helping the T–F axis increase Accuracy by $\sim 10\%$.

+ Improved Balance: Macro F1 surged (N–S: $59\% \rightarrow 71\%$, I–E: $62\% \rightarrow 72\%$), confirming that the filters learned minority group features instead of rote memorization of the majority.

+ Stability: More consistent growth than LSTM across all four axes; however, overall performance remains lower than LinearSVC, requiring deeper fine-tuning to surpass the linear model.

4.1.3 Transformer-based Models (RoBERTa Fine-tuning)

Trục (Axis)	Bộ Dữ Liệu	Accuracy (Độ chính xác)	Macro Avg F1 (Độ cân bằng)	Weighted Avg F1 (Tổng quan)
IE	mbti_1	77.75%	67.00%	78.00%
	MBTI-500	80.52%	74.72%	81.03%
NS	mbti_1	74.93%	58.00%	77.00%
	MBTI-500	88.88%	72.52%	90.04%
TF	mbti_1	85.42%	85.00%	85.00%
	MBTI-500	85.25%	84.30%	85.47%
JP	mbti_1	55.00%	59.00%	53.00%
	MBTI-500	75.13%	74.72%	75.24%

Figure 8: Evaluation of RoBERTa Model Performance

+ Effort to Explore SOTA: Advanced experiments aimed at leveraging the Transformer architecture to capture deep semantics, going beyond mere lexical statistics.

+ Improvement on the J–P Axis with Large Data: The most significant breakthrough, raising performance from "guessing" (55%) to a usable level ($\sim 75\%$), proving that Transformers require large-scale data to converge.

+ Breakthrough on the N–S Axis: Dramatic growth (+14% Accuracy), affirming the ability to grasp abstract reasoning when sufficient resources are provided.

+ Conclusion: Although fairness (Macro F1) improved markedly and demonstrated great potential, overall RoBERTa still did not surpass the performance of classical Baseline models, indicating that the complexity of SOTA is not necessarily more optimal than the simplicity of statistical approaches on this dataset.

4.2 Comparison of Models

Based on the experimental results, the team conducted a comparative analysis to clarify the advantages and disadvantages of each approach.

Trục	Mô hình (Model)	Accuracy (mbti_1)	Accuracy (MBTI-500)	Macro F1 (mbti_1)	Macro F1 (MBTI-500)	Weighted F1 (mbti_1)	Weighted F1 (MBTI-500)
IE	LinearSVC (SVM)	0,8167	0,8427	0,69	0,8006	0,8	0,8484
	Logistic Regression	0,7712	0,8845	0,68	0,84	0,77	0,89
	LSTM	0,7205	0,7953	0,65	0,729	0,73	0,7916
	TextCNN	0,7671	0,79	0,62	0,7198	0,75	0,7929
	RoBERTa	0,7775	0,8052	0,67	0,7472	0,78	0,8103
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NS	LinearSVC (SVM)	0,8732	0,9134	0,6	0,7789	0,84	0,9214
	Logistic Regression	0,8576	0,9114	0,71	0,79	0,86	0,92
	LSTM	0,849	0,8146	0,56	0,622	0,82	0,8461
	TextCNN	0,8427	0,8899	0,59	0,7103	0,82	0,8988
	RoBERTa	0,7493	0,8888	0,58	0,7252	0,77	0,9004
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TF	LinearSVC (SVM)	0,7965	0,8947	0,8	0,8856	0,8	0,8955
	Logistic Regression	0,796	0,9281	0,8	0,86	0,8	0,93
	LSTM	0,728	0,8608	0,71	0,7377	0,71	0,872
	TextCNN	0,7291	0,828	0,73	0,8146	0,73	0,8298
	RoBERTa	0,8542	0,8525	0,85	0,843	0,85	0,8547
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JP	LinearSVC (SVM)	0,7084	0,7972	0,69	0,7933	0,7	0,7979
	Logistic Regression	0,7078	0,8217	0,7	0,82	0,71	0,82
	LSTM	0,6121	0,7197	0,41	0,6975	0,48	0,7148
	TextCNN	0,6674	0,7008	0,64	0,6965	0,66	0,7023
	RoBERTa	0,55	0,7513	0,59	0,7472	0,53	0,7524

Figure 9: Comparative Performance Results of Models on Two Datasets

4.2.1 Effectiveness of Data Expansion (Dataset Impact)

Accuracy: Transitioning from mbti_1 to MBTI-500 improved performance by an average of 10% to 20% across all models.

Macro F1: The Macro F1 score on the new dataset surged and approached Accuracy (with only a small gap), whereas on the old dataset, Macro F1 was often 15–20% lower than Accuracy. This indi-

cates a clear improvement in recognizing minority groups.

4.2.2 Model Ranking

Baseline Group (Logistic/SVM): Maintained Rank 1 across both datasets, affirming the absolute effectiveness and stability of statistical methods for the characteristics of this data.

RoBERTa (Transformer): Recorded the most significant progress, rising from the bottom group on mbti_1 (due to lack of data) to Rank 2 on MBTI-500, proving that this architecture requires large-scale data to be effective.

Deep Learning (LSTM/CNN): Although absolute performance improved, their relative ranking was pushed down to the lower group (Ranks 3 & 4) because they could not keep pace with RoBERTa's strong growth and the stability of the Baseline.

4.2.3 Axis Comparison

N–S Axis (Intuition – Sensing): This axis achieved the highest accuracy across all models (>91% with LinearSVC).

J–P Axis (Judging – Perceiving): This axis had the lowest accuracy (10–15% lower than N–S). However, it showed the greatest growth margin when moving to the new dataset (from $\sim$ 55–60% up to $\sim$ 80%).

T–F & I–E Axes (Thinking – Feeling & Introversion – Extraversion): Maintained fairly stable performance (80%–92%), with the T–F axis showing the strongest breakthrough in the Logistic Regression model.

5 Discussion and Evaluation

Based on the comparative experimental results between the two datasets (mbti_1 vs MBTI-500) and the three model groups (Baseline, Deep Learning, Transformer), this chapter delves into the interesting phenomena observed, particularly the "performance paradox" between traditional and modern models.

5.1 The "Big Data" Effect

The experimental results affirm a core truth in Data Science: data determines the performance ceiling of a model.

- Qualitative Transformation: Expanding the dataset from 8.6k samples to 106k samples increased the average performance of all models by 10%–20%.

- Stabilizing Deep Learning: On the smaller mbti_1 dataset, models such as LSTM and TextCNN fluctuated heavily and frequently fell into "Mode Collapse" (predicting entirely the majority class). With MBTI-500, these models converged stably. This demonstrates that complex neural architectures necessarily require large-scale data to avoid overfitting.

5.2 Decoding the "Performance Paradox": Why Do Baseline Models Outperform SOTA?

An important finding is that the baseline model (TF-IDF + Logistic Regression/SVM) achieved superior results compared to the SOTA model (RoBERTa). The research team identified three core reasons based on the characteristics of the task:

- Advantage of Lexical Features: Personality on social media is often revealed through topical keywords (e.g., "party" vs "alone") rather than complex grammatical structures. TF-IDF directly captures these frequency signals. In contrast, RoBERTa attempts to analyze grammatical structures that are noisy (slang, misspellings) in Reddit data, which leads to "overcomplicating" the problem and learning spurious signals (Noise).

- Global Context Window: RoBERTa is limited to 512 tokens of input, forcing truncation of text and resulting in information loss. Meanwhile, TF-IDF can scan the entire text regardless of length, ensuring that no important keywords are missed and providing a more comprehensive overview of the user.

- Linear Separability: In the 10,000-dimensional feature space generated by TF-IDF, data points tend to be linearly separable. Thus, a simple hyperplane from LinearSVC is sufficient for accurate classification. Applying Deep Learning to warp the feature space in this case is redundant (Overkill) and prone to overfitting.

5.3 Analysis of the Causes of Performance Variation across Axes

Experimental results reveal a clear differentiation in performance across personality axes, reflecting their varying degrees of manifestation in text. Specifically, the N–S (Intuition – Sensing) and T–F (Thinking – Feeling) axes achieved the highest accuracy (over 91% across all models) thanks to strong semantic separability at the lexical level. While the N–S axis shows a sharp contrast between

abstract language (theory, dream) and concrete language (job, table), the T–F axis exhibits strong polarization between emotional words (feel, sad) and logical reasoning words (analyze, because), creating "anchor signals" that allow models to easily establish accurate classification boundaries.

In contrast, the J–P (Judging – Perceiving) and I–E (Introversion – Extraversion) axes became "low-performance zones" due to behavioral ambiguity. Particularly for J–P, both groups often use the same vocabulary (e.g., "deadline", "plan") but differ in their attitudes toward execution; this semantic overlap makes it difficult for statistical models to distinguish between them. Similarly, the I–E axis is heavily influenced by online noise, where introverts tend to write more actively and enthusiastically on anonymous forums, blurring the boundaries of communicative energy compared to real-life contexts.

5.4 Limitations

Although the results are promising, the study still faces core data-related limitations that affect generalizability. First is the reliability of self-reported labels, which introduces label noise since users often misperceive their own personality. Next is the severe issue of data imbalance caused by self-selection bias, with introverts and intuitives overwhelmingly represented on forums, making it difficult to identify minority groups despite technical handling efforts. Finally, the dataset carries strong community bias, being dominated by young, tech-savvy Reddit users, which limits the applicability of the models to broader demographic groups.

From a technical perspective, the study notes a disproportionate trade-off between cost and effectiveness; deploying RoBERTa consumed computational resources dozens of times greater than Logistic Regression, yet the performance improvement was not significantly superior in this context. Most critically, hardware infrastructure limitations posed the greatest barrier to advanced models. Due to insufficient computational resources (GPU memory), the team was forced to truncate input text length and restrict training parameters, preventing Deep Learning and RoBERTa architectures from fully capturing the deep context of entire posting histories, thereby limiting their ability to realize their theoretical potential.

6 Conclusion and Future Directions

6.1 Summary of the Study

This research conducted a comprehensive and systematic investigation into the problem of predicting MBTI personality types based on social media text data. By constructing two parallel datasets — mbti_1 (8.6k samples) and MBTI-500 (106k samples) — and implementing a diverse range of architectures from traditional Machine Learning (Logistic Regression, SVM) to Deep Learning (LSTM, CNN) and Transformer (RoBERTa), the project successfully addressed the multi-label classification task in the context of noisy and imbalanced data.

6.2 Contributions of the Study

The contributions of this research are comprehensively reflected across three core dimensions. From a scientific perspective, the study empirically demonstrated the decisive role of the "Data-Centric AI" paradigm, confirming that expanding dataset scale is the key factor to thoroughly overcoming overfitting and achieving superior performance (10–20%) across all model architectures. From a linguistic perspective, the research clarified the nature of the MBTI task, showing that personality on social media is primarily revealed through lexical frequency (keywords) rather than grammatical structures, thereby establishing the optimality of baseline statistical models (Logistic Regression) in terms of the performance–cost balance, while also highlighting the irreplaceable role of SOTA models (RoBERTa) in addressing latent behavioral traits such as J–P. Finally, from a practical perspective, the study proposed a standardized pipeline — from eliminating data leakage to model selection — laying the foundation for feasible applications in automated recruitment systems and empathetic marketing, with low cost and high reliability.

6.3 Future Directions

To overcome current limitations and improve accuracy, the team proposes the following directions:

- First, expansion to multimodal and multi-scale analysis: Combine text with non-linguistic behavioral data (posting time, interactions) to resolve ambiguity in the J–P axis, while experimenting with universal personality models such as the Big Five (OCEAN) to address the rigid binary limitation of MBTI.

- Second, improving data quality: Replace self-reported labels with standardized psychological tests to eliminate noise.

- Finally, leveraging next-generation language models with long-context windows (Long-context LLMs): When hardware conditions permit, process entire posting histories without truncation, thereby fully capturing deep contextual information.

7 Reference

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